



Security Assessment Report



Etherfi – EtherFiSpokeInstance

July 2026

Prepared for EtherFi

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Project Summary

Project Scope

Project Name	Repository Link	Commit Hash	Latest Commit Hash	Platform
Etherfi – EtherFiSpokeInstance	https://github.com/etherfi-protocol/cash-v3/pull/209	9838962a712a6197ebe3cfde40bc6500e8b90607	9838962a712a6197ebe3cfde40bc6500e8b90607	Solidity

Project Overview

This document describes the security review of **Etherfi – EtherFiSpokeInstance**. The work was undertaken on **July 30th, 2026**.

The following contract list is included in our scope:

- src/aave-v4/EtherFiSpokeInstance.sol

Protocol Overview

The [etherfi-protocol/cash-v3](#) repository contains the smart contracts for **ether.fi Cash**, a crypto-native banking alternative designed to consolidate how users save, grow, and spend their digital assets. Built on decentralized infrastructure, it provides users with true self-custody through multi-signature smart contract wallets rather than relying on centralized third-party custodians. The platform enables programmable finance and censorship-resistant spending rails while natively offering competitive cashback and DeFi yields. Core modular components include



EtherFiSafe for account security, **CashModule** for spending capabilities, and **DebtManager** for transparent on-chain lending and borrowing.

Assessment Methodology

Our assessment approach combines design level analysis with a deep review of the implementation to ensure that a protocol is secure, economically sound, and behaves as intended under realistic conditions.

At the design level, we evaluate the architecture, the economic assumptions behind the protocol, and the safety properties that should hold independently of a specific chain or environment. This process includes reviewing internal and cross protocol interactions, state transition flows, trust boundaries, and any mechanism that could be exploited to extract value, deny service, or alter core system behavior. At this stage, a focused threat modelling exercise helps identify key attack surfaces and adversarial capabilities relevant to the system. Design level issues often relate to incentive structures, governance implications, or systemic behavior that emerges under adversarial conditions.

Implementation analysis focuses on the concrete behavior of the code within the execution model of the target chain. This involves reviewing the correctness of logic, access control, state handling, arithmetic behavior, and the nuanced behaviors of the chain environment. Familiar classes of vulnerabilities such as reentrancy conditions, faulty permission checks, precision issues, or unsafe assumptions often surface at this layer. These findings require context aware reasoning that takes into account both the code and the architectural intent.

To support this analysis, the codebase is examined through repeated manual passes and supplemented by automated tools when appropriate. High-risk logic areas receive deeper scrutiny, invariants are validated against both design intent and actual implementation, and potential vulnerability leads are thoroughly investigated. Automated techniques such as static analysis, fuzzing, or symbolic execution may be used to complement manual review and provide additional insight.

Collaboration with the development team plays an important role throughout the audit. This helps confirm expected behaviors, clarify design assumptions, and ensure an accurate understanding of the protocol's intended operation. All findings are documented with clear reasoning, reproducible examples, and actionable recommendations. A follow up review is conducted to validate the applied fixes and verify that no regressions or secondary issues have been introduced.

Findings Summary

The table below summarizes the findings of the review, including type and severity details.

Severity	Discovered	Confirmed	Fixed
Critical	-	-	-
High	-	-	-
Medium	-	-	-
Low	-	-	-
Informational	-	-	-
Total	-	-	-

Severity Matrix

Impact	High	Medium	High	Critical
	Medium	Low	Medium	High
	Low	Low	Low	Medium
		Low	Medium	High
Likelihood				

Detailed Findings

ID	Title	Severity	Status
-	-	-	-

No issues were reported during the security assesment of this scope

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About Certora

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Certora also provides services such as auditing, formal verification projects, and incident response.